Graefe's Archive for Clinical and Experimental Ophthalmology (2018) 256:1299–1304
https://doi.org/10.1007/s00417-018-3988-7

BASIC SCIENCE

Effect of Pycnogenol® on an experimental rat model of allergic conjunctivitis

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Received: 13 November 2017 / Revised: 15 February 2018 / Accepted: 11 April 2018 / Published online: 20 April 2018

Abstract

Background Allergic conjunctivitis (AC) is a frequent and challenging disease in ophthalmology practice. Cell protective effect of Pycnogenol® (PYC) depends on its antioxidant and anti-inflammatory properties. The aim of the study is to investigate the effect of PYC on an experimental AC model.

Methods Ovalbumin and $Al(OH)_3$ were given seven times intraperitoneally (i.p.) every other day and ovalbumin installed everyday directly on conjunctiva to create an AC rat model. Then, PYC (3 or 10 mg/kg i.p.) was applied in the study groups. Control rats were given adjuvant $Al(OH)_3$ i.p. and topical saline on conjunctiva. A negative control group in which only PYC (10 mg/kg/7 days) was administered i.p. and an AC positive control group which have been given dexamethasone (1 mg/kg/7 days) was created. Mast cells were counted with a microscope; histological evaluation was performed with H-E and toluidine blue, mast cell tryptase, and TNF-α and TGF-β staining.

Results Pycnogenol treatment alone did not show any detrimental effect. Mast cell count (MCC) decreased in both dexamethasone and 10 mg/kg given PYC treatment groups compared to positive control group and these results were statistically significant (MCC 1.85 ± 0.69, $p < 0.001$; 2.42 ± 0.53, $p = 0.003$). Negative staining with TGF-β and weak focal staining with TNF-α were the common findings of dexamethasone and PYC treatment groups.

Conclusions The animal model of AC was successfully developed by using aforementioned way. PYC is a safe herbal product and it has alleviated the findings of ovalbumin-induced AC—similar to dexamethasone—histologically in this experimental model. These results are promising for the future of AC treatment.

Keywords Allergy · Conjunctivitis · French maritime pine · Pycnogenol® · Rat

This study has been presented as a free paper at ESCRS 2016, Copenhagen, Denmark.

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Introduction

Allergic conjunctivitis (AC) is a frequent and challenging disease in ophthalmology practice. It has a wide and changing spectrum of clinical forms from mild seasonal allergic conjunctivitis to vernal keratoconjunctivitis. It is an immunoglobulin E (IgE)-mediated type 1 hypersensitivity reaction and mast cells; T helper-2 lymphocytes and eosinophils play the major roles by secreting histamine, cytokines, and chemokines [1, 2]. Due to histamine and inflammatory mediator discharge, congestion, ciliary injection, edema, watery discharge, pruritus, photophobia, and blurred vision occur and the quality of life is affected significantly. Also, cytokines like TNF-α destroy the tissues and cause permanent damage. Overall, allergic conjunctivitis could be a sight-threatening problem by limbal stem cell deficiency, corneal neovascularization, pannus, and keratoconus formation [3–6]. If the increasing number of patients with AC and its sight-threatening complications is taken into account, treatment of AC is essential [7, 8].

Although safe and effective drugs are available such as anti-histamines, mast cell stabilizers, and NSAI drugs for mild cases, severe ones may still need corticosteroid treatment. Cyclosporine A is another option but it is not as effective as steroids. Although steroid treatment is highly effective, it may cause several serious side effects such as induction of cataract, glaucoma, and infections. Therefore, new agents are needed which should be effective as corticosteroids without its abovementioned side effects. The natural plant derivatives are gaining popularity as a new way of remedies. Pycnogenol (PYC) is one of these natural plant derivatives with its antioxidant and anti-inflammatory effect. It is a standardized extract of the French maritime pine bark (*Pinus maritime*). It contains procyanidins, bioflavonoids, and phenolic acids. It is used for supporting healthy aging, decreasing blood cholesterol and pressure [9–12].

Studies concerning the effect of PYC on IgE levels were the mainstay of this study about the idea of PYC trial on allergic conjunctivitis [13–15]. Therefore, the aim of this study was to determine the safety of PYC extract on eye histolopathology and to investigate the efficacy of PYC on an ovalbumin-induced AC model. Additionally, one of the allergic rhinitis rat models has been modified to see whether it is a suitable rat model of AC or not.

Materials and methods

Animals

Local Animal Ethical Committee's approval was obtained (HADYEK 2014/098). Female Wistar rats aged between 12 and 15 months were obtained from the Experimental Animal Center of the same institution. The rats were randomly divided into six groups, with each cage containing seven rats.

Sensitization design

In order to sensitize the animals, 30 mg/mL adjuvant $Al(OH)_3$ (Sigma) and 0.3 mg ovalbumin (OVA; Sigma) in 1 mL saline have been applied intraperitoneally (i.p.) every other day for seven times. Consequently, 2% OVA-saline solution was dropped into both conjunctiva once a day, for 7 days to establish an AC animal model. Meanwhile, saline, dexamethasone, and PYC treatments were given i.p. after the establishment of the AC model. Actually, this model was modified from two previous researches about allergic rhinitis model [16, 17].

Experimental design

Six groups were conducted for the study. The first group was the control/sham group. It was free from OVA; only 30 mg/mL adjuvant $Al(OH)_3$ in 1 mL saline i.p. and 7 days of saline drops on both eyes have been applied. The second group was the AC group, showing the establishment of the AC model. It was designed as it is told in the sensitization design section. The third group was the positive control group containing the rats treated with dexamethasone. Dexamethasone (Dekort® amp) 1 mg/kg was given i.p. to the AC group similar to PYC intervention. Fourth and fifth groups were the PYC (Horpag Research, Geneva, Switzerland) treatment groups. PYC was applied either 3 or 10 mg/kg i.p. in 1 mL saline between 14th and 21st days of the study seven times along with the OVA eye drops. By this way, the dose-dependent treatment effect could be evaluated. Additionally, the sixth group was the negative control group containing 10 mg/kg PYC® injection to reveal the safety of agent in these dose, course, and route. At the end of the study, under the anesthesia of ketamine and xylazine (5 and 50 mg/kg, respectively), rats were harvested and the globes have been exenterated together with the eyelids.

Histopathological examination

For histopathological examination, specimens were placed in formalin and sent to the pathology laboratory for routine processing and hematoxylin-eosin (HE) and toluidine blue (TB) staining. Histopathological examination was made under a light microscope (BX51, Olympus, Tokyo, Japan) with × 10, × 20, and × 40 magnification. The number of mast cells was counted at three high power (× 400) consecutive or nonoverlapping fields by using a light microscope. Mast cells were quantified in the foci of densest field at × 400 power and reported as absolute number per high power field (HPF).

Immunohistochemistry

The immunostaining was carried out at room temperature using the DAKO Autostainer Universal Staining System (Autostainer Link 48 DAKO, Glostrup, Denmark). First, sections 4 μm in thickness obtained from selected paraffin-embedded blocks were put on positively charged slides. Second, all the sections were deparaffinized in xylene and dehydrated through a graded series of ethanol solution. Third, antigen retrieval was performed at 96 °C (10 mM/L citrate buffer, pH 6) for 40 min in a thermostatic bath (PT Link). The sections were incubated with anti-mast cell tryptase (MCT) (prod. no: M705229, DAKO, Glostrup, Denmark), anti-TNF-α (prod. no: ab9739, Abcam, Cambridge, MA, USA), and antiTGF-β (prod. no: ab92486, Abcam, Cambridge, MA, USA) for 60 min at room temperature. Positive and negative controls were added for antibody. A streptavidin-biotin-enhanced immunoperoxidase technique (K8000 Envision Flex, DAKO, Glostrup, Denmark) in an automated system was used to show immunoreactions. The sections were incubated with DAB and counterstained lightly with hematoxylin to demonstrate binding. Finally, the sections were dehydrated and mounted onto the slides. The slides known to be positively immunostained were used as positive controls. Normal rabbit serum IgG was used to replace primary antibody as a negative control.

Evaluation of immunostaining

Immunostaining was scored (as described by Zhigang) [18] based on the intensity of staining and the percentage of cells that were stained positively. The intensity of expression was graded on a scale of 0 to 3+ with 3 being the highest expression observed (0, no staining; 1+, mildly intense; 2+, moderately intense; 3+, severely intense). The staining density was quantified as the percentage of cells stained positively for immunohistochemical antibodies, as follows: 0 = no staining, 1 = positive staining in < 25% of the sample, 2 = positive staining in 25–50% of the sample, and 3 = positive staining in > 50% of the sample. Intensity score (0–3+) was multiplied by the density score (0–3) to yield an overall score of 0–9 for each specimen, i.e., 0, negative expression; 1–2, weak expression; 3–6, moderate expression; and 7–9, strong expression.

Statistics

Comparison between six groups was made by Kruskal-Wallis test and descriptive statistics are presented as mean ± standard deviation. The p values below 0.05 were considered statistically significant.

Results

Average mast cell count (MCC) of the constructed AC model (6.43 ± 0.98) was significantly higher than that of the control (1.43 ± 0.53) and PYC challenge (2.57 ± 0.53) groups, as well as the three treatment groups (Table 1). Mast cell count was significantly decreased by both 1 mg/kg dexamethasone (1.85 ± 0.69), 10 mg/kg PYC (2.42 ± 0.53), and 3 mg/kg PYC (2.88 ± 0.64) treatment groups on the experimental model of AC ($p < 0.001$, $p = 0.003$, and $p = 0.027$, respectively). Comparison of the treatment groups and the control group revealed no statistically significant difference for MCC, except for the slight difference found between the 3 mg/kg PYC treatment group and controls ($p = 0.004$). No statistically significant difference was found between average MCC of control and PYC challenge groups. All the results are summarized in Table 1.

In the same manner, pathological evaluation of MCT staining of the AC group was remarkable; on the other side, treatment groups containing dexamethasone and PYC were negatively stained. Weak focal staining of TNF-α was the common findings of the treatment groups. Inversely, TGF-β staining revealed the destruction of the conjunctiva epithelium on the AC group and the integrity of the tissue on the treatment groups. PYC application at 10 mg/kg dosage alone did not show any detrimental effect on the histopathology of conjunctiva (Fig. 1).

Discussion

Allergic conjunctivitis is a disturbing disease with its itching, burning, and tearing sensation. With these symptoms, AC decreases the quality of life and also threatens the vision with its complications. As the environmental pollution increases, the prevalence of the AC increases and clinicians are seeking for new treatment alternatives instead of steroids which have the risk of cataract, glaucoma, and infection [3–8].

Establishing an experimental AC model is important in order to develop new treatment strategies. To develop the AC model in this study, two different researchers' replicable rat rhinitis models were modified [16, 17]. Both researchers dropped OVA into the nasal mucosa for 7 days after

Table 1 Post-hoc comparison of study groups according to mast cell counts

Compared groups	First variable MCC ± SD	Second variable MCC ± SD	p
Control vs. AC	1.43 ± 0.53	6.43 ± 0.98	< 0.001
Control vs. PYC	1.43 ± 0.53	2.57 ± 0.53	0.352
Control vs. PYC3AC	1.43 ± 0.53	2.88 ± 0.64	0.004
Control vs. PYC10AC	1.43 ± 0.53	2.43 ± 0.53	0.385
Control vs. CSAC	1.43 ± 0.53	1.85 ± 0.69	0.417
AC vs. PYC	6.43 ± 0.98	2.57 ± 0.53	0.007
AC vs. PYC3AC	6.43 ± 0.98	2.88 ± 0.64	0.027
AC vs. PYC10AC	6.43 ± 0.98	2.42 ± 0.53	0.003
AC vs. CSAC	6.43 ± 0.98	1.85 ± 0.69	< 0.001
PYC vs. PYC3AC	2.57 ± 0.53	2.88 ± 0.64	0.578
PYC vs. PYC10AC	2.57 ± 0.53	2.43 ± 0.53	0.747
PYC vs. CSAC	2.57 ± 0.53	1.85 ± 0.69	0.146
PYC3AC vs. PYC10AC	2.88 ± 0.64	2.43 ± 0.53	0.374
PYC3AC vs. CSAC	2.88 ± 0.64	1.85 ± 0.69	0.592
PYC10AC vs. CSAC	2.43 ± 0.53	1.85 ± 0.69	0.257

AC allergic conjunctivitis, *PYC* Pycnogenol, *PYC3AC* AC model treated with 3 mg/kg PYC, *PYC10AC* AC model treated with 10 mg/kg PYC, *CSAC* AC model treated with 1 mg/kg dexamethasone

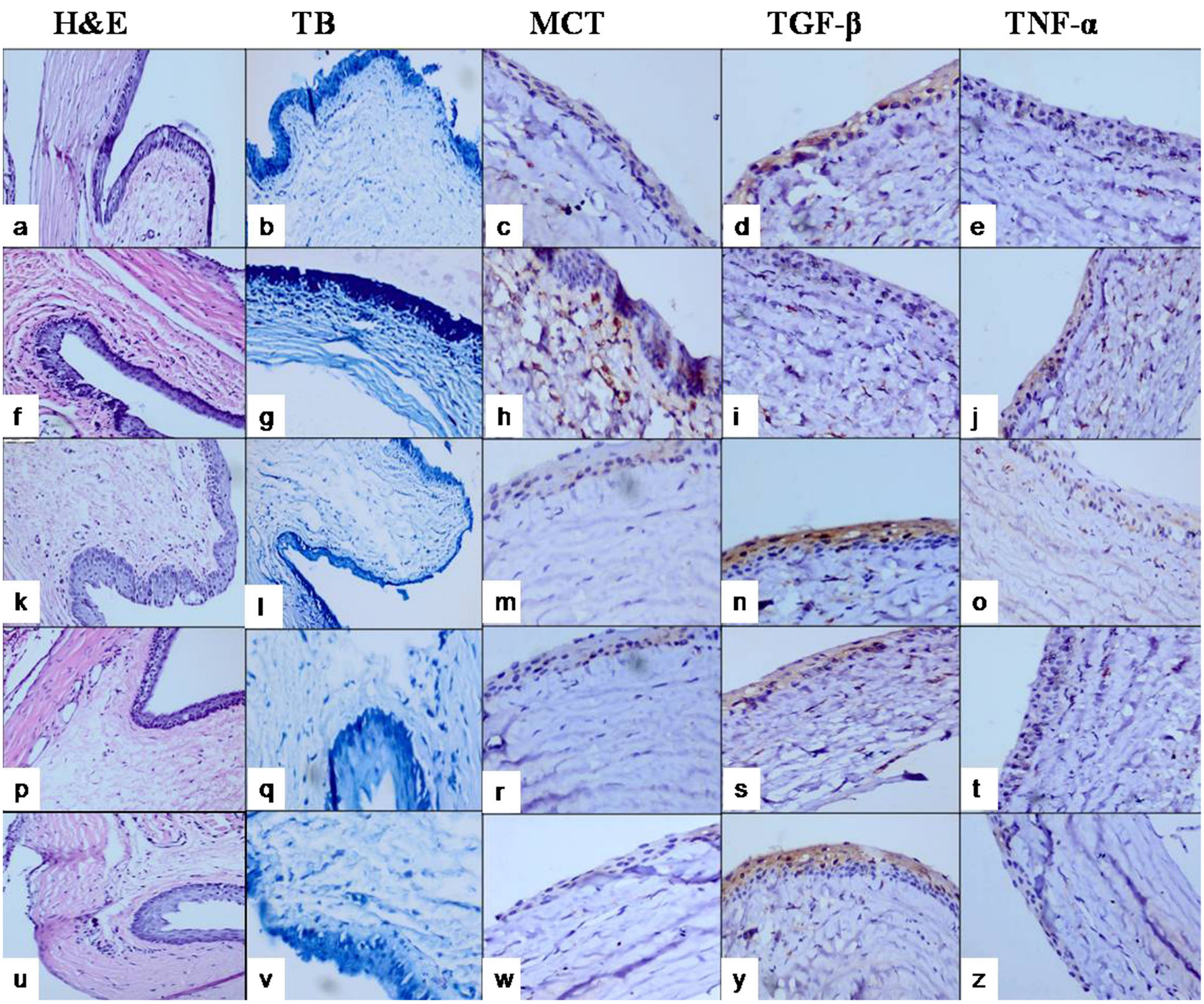

Fig. 1 Pathological staining patterns of all study groups. **a**–**e** Control group. **f**–**j** Allergic conjunctivitis group. **k**–**o** Dexamethasone treatment group. **p**–**t** Pycnogenol treatment group. **u**–**z** Pycnogenol challenge group. H&E hematoxylin-eosin, TB toluidine blue, MCT mast cell tryptase, TGF-β transforming growth factor-β, TNF-α tumor necrosis factor-α

sensitization period; instead, the OVA-saline solution has been dropped on both eyes for 7 days following sensitization. As the previous researchers' model was easy and the rats were the only available animals in our institution, we could not mimic the well-known mouse allergic conjunctivitis models in the literature [19, 20]. According to the results of MCC and histological evaluation, this modification was found to be an easy and successful way of developing an AC model of rats.

The popularity of natural plant derivatives is increasing as a new way of remedies. The first study about PYC's beneficial effect on ocular tissue has been done in 1996; it has been reported that application of 10^{-5} M has shown partially antioxidant effect on the model of lipid peroxidation produced by ferric iron on bovine and porcine retinal layers [21]. Similarly, when given i.p. 10 mg/mL/day for 14 days, it balanced antioxidant system on retinal tissue of streptozosin-induced diabetic rats [22].

One of the current and remarkable studies about the safety of PYC is its comparison with aspirin on recurrence of retinal vein thrombosis [23]. PYC seems to be effective with its anti-inflammatory, anti-edema, and anti-platelet aspects while revealing its safety profile.

Taner et al. have proven PYC's decreasing effect on tumor necrosis factor-alpha (TNF-α) level in septic rats [11]. This study also supports our finding that PYC modulates TNF-α level on eye tissue as it has been shown immunohistologically. Additionally, we have shown that PYC protected the conjunctiva epithelium on an allergic condition at the TGF-β staining and clearly proven that PYC had anti-inflammatory effects based on cytokine pathway.

Previous studies investigating the effect of PYC on allergic asthma and rhinitis also support the results of the presented study. Wilson et al. had reported lower scores of eye and nasal symptoms of patients with allergic rhinitis. This Canadian study with 39 subjects has shown that birch-specific IgE levels increased 31.9% in the placebo group while the PYC group only had 19.4% of IgE in allergic season [14]. MCC decreased 55.2% in the 3 mg/kg PYC group and 62.3% in the 10 mg/kg PYC group compared to the AC group in the presented study. Shin et al. investigated the effect of PYC on an OVA-induced experimental allergic asthma model and stated that PYC decreases nitric oxide synthetase, metalloproteinase-9, inflammatory cell count, and IL-4, IL-5, IL-13, and IgE levels and increases the expression of hemeoxygenase-1 [13]. An experimental OVA-induced allergic rhinitis study also revealed that serum IgE, IL-4, IL-10, and IFN-γ levels and IL-1β expression on tissues decreased significantly with PYC treatment [15]. Lower MCC, weak staining of TNF-α, and normal TGF-β staining of the PYC treatment groups show the anti-allergic effect of PYC similarly to the above studies.

One important limitation of the presented study is the route which PYC has been given. As well known, the agents' distribution and uptake of the tissue have some limitations such as absorption capacity or metabolism on the liver. Additionally, the penetration of the eye can be varied due to the barrier of ocular tissues, so that the way of treatments has to be free from unpleasant side effects and is worth investigating about the effectiveness on the disease. It should be considered that most of agents, drugs, and foods produce allergic reactions on humans. Some of them, including PYC, have been taken on systematical route as a food supplement. Therefore, PYC is chosen to given systematically in this study instead of using it on local eye drops to investigate its safety.

Although PYC has several negative sides, this study has revealed that PYC itself did not increase the mast cell count or trigger TNF-α or alter TGF-β which are the markers of allergic reactions and inflammation. Pathological findings showing the reduced number of mast cells, MCT, and especially proinflammatory cytokine TNF-α staining patterns indicate the safety and effectivity of PYC on AC. Also, TGF-β staining reveals that PYC protects the tissue from the destruction of inflammatory process of AC similar to the steroids.

As a conclusion, after 14 days of sensitization period, OVA installation on conjunctivae is a suitable model for AC studies. Seven days treatment with 10 mg/kg of PYC is a safe and effective treatment of AC even if it is taken systematically. These results suggest that PYC can become a promising treatment option for AC patients. Further experimental and clinical studies are necessary to carry Pycnogenol from bench to bedside.

Acknowledgments We wish to thank to Horpag Research Ltd. for generously gifting us with PYCnogenol®.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical statement All applicable international, national, and/or institutional guidelines for the care and use of animals were followed. All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted.

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